

Discrete Mathematics Spring 2026

Homework 8: Functions

Due Sunday April 5 @11:59:00pm Eastern

Show all your work to receive full credit

1. Function (definitions)

Let the set $A = \{1, 2, 3, 4, 5, 6\}$. Let's define the function h on the power set of A , $\mathcal{P}(A)$, as follows:

$$h : \mathcal{P}(A) \rightarrow \mathbb{Z}$$

defined by

$$h(X) = \text{sum of all elements in } X$$

(Note: $h(\emptyset) = 0$ since the empty sum is 0)

1. What is the domain of h ? What is the codomain of h ?
2. Calculate $h(\{2, 5\})$ and $h(\{1, 3, 4\})$.
3. What is the pre-image of 7 under h ?
4. What is the image of h ?
5. Is h one-to-one? Prove or provide a counter-example.
6. Is h onto? Prove or provide a counter-example.

2. Composition of functions

For each of the following pairs of functions f and g , calculate $g \circ f$ and $f \circ g$

1. $f = \{(1, 2), (3, 4), (5, 6)\}$ and $g = \{(2, 1), (4, 3), (6, 5)\}$
2. $f(x) = 2x + 1$ and $g(x) = 4x - 5$ for all $x \in \mathbb{Z}$

3. Bijection, inverse functions and composition

Consider two functions:

$$f : \mathbb{Z} \rightarrow \mathbb{Z} \text{ defined by } f(n) = 2n + 1$$

$$g : \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } g(x) = 2x + 1$$

- (a) Prove that f is not onto.
- (b) Prove that g is a bijection.
- (c) Find the inverse of g .
- (d) Calculate $g^{-1} \circ g$ and $g \circ g^{-1}$.

- (e) What can you say about the result of composing a function with its inverse ($g \circ g^{-1}$) and composing the inverse with the original function ($g^{-1} \circ g$)?

4. **Onto, one-to-one, bijection**

Consider the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by:

$$f(n) = \begin{cases} n + 1 & \text{if } n \equiv 0 \pmod{3} \quad (\text{remainder 0 when divided by 3}) \\ n - 1 & \text{if } n \equiv 1 \pmod{3} \quad (\text{remainder 1 when divided by 3}) \\ n & \text{if } n \equiv 2 \pmod{3} \quad (\text{remainder 2 when divided by 3}) \end{cases}$$

For example: $f(3) = 4$, $f(4) = 3$, $f(5) = 5$, $f(6) = 7$, $f(7) = 6$, $f(8) = 8$.

1. Is f one-to-one? If yes prove it, otherwise provide a counter-example.
2. Is f onto? If yes prove it, otherwise provide a counter-example.
3. Is f a bijection? If so, find f^{-1} .

5. **Pigeonhole principle**

Prove that if seven distinct numbers are selected from 2, 3, 4, . . . , 12, then some two of these numbers will add up to 14. (you can assume that the 7 can be picked up more than once).

Identify the pigeons and the holes. Then, using the Pigeonhole Principle, prove that the statement is true.